(3.6) Theorem. *Assume the diagram of G-modules and G-homomorphisms*

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A' \arrow[r] \arrow[d] & A \arrow[r] \arrow[d] & A'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & B' \arrow[r] \arrow[d] & B \arrow[r] \arrow[d] & B'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & C' \arrow[r] \arrow[d] & C \arrow[r] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

is commutative with exact rows and columns. Then the diagram

$$\begin{tikzcd}
H^{q-1}(G,C'') \arrow[r, "\delta"] \arrow[d, "\delta"] & H^q(G,C') \arrow[d, "-\delta"] \\
H^q(G,A'') \arrow[r, "\delta"] & H^{q+1}(G,A')
\end{tikzcd}$$

commutes.

Proof. Let D be the kernel of the composite map $B \to C''$; thus the sequence

$$0 \longrightarrow D \longrightarrow B \longrightarrow C'' \longrightarrow 0$$

is exact. We define G-homomorphisms

$i : A' \to A \oplus B'$ by $i(a') = (a, b')$, where a (resp. b') is the image of a' in A (resp. of a' in B'),

$j : A \oplus B' \to D$ by $j(a, b') = d_1 - d_2$, where d_1 (resp. d_2) is the image of a (resp. of b') in $D \subseteq B$.

It is easy to verify that with these definitions the sequence

$$0 \longrightarrow A' \xrightarrow{i} A \oplus B' \xrightarrow{j} D \longrightarrow 0$$

is exact, and that the diagram

$$\begin{tikzcd}
A' \arrow[r] & A \arrow[r] & A'' \arrow[r] & B'' \arrow[r] & C'' \\
A' \arrow[u, equal, "\mathrm{id}"] \arrow[d, equal, "-\mathrm{id}"'] \arrow[r, "i"] & A \oplus B' \arrow[u, "{(\mathrm{id},0)}"'] \arrow[d, "{(0,-\mathrm{id})}"] \arrow[r, "j"] & D \arrow[u, dashed] \arrow[d, dashed] \arrow[r] & B \arrow[u] \arrow[d] \arrow[r] & C'' \arrow[u, equal, "\mathrm{id}"'] \arrow[d, equal, "\mathrm{id}"] \\
A' \arrow[r] & B' \arrow[r] & C' \arrow[r] & C \arrow[r] & C''
\end{tikzcd}$$

commutes. Because $\mathrm{im}\,(D \to B'') \subseteq \mathrm{im}\,(A'' \to B'')$ and $A'' \to B''$ is injective, there is a G-homomorphism $D \to A''$ which extends the above diagram. Similarly, since $\mathrm{im}\,(D \to C) \subseteq \mathrm{im}\,(C' \to C)$ and $C' \to C$ is injective, there is an analogous G-homomorphism $D \to C'$. Since the resulting extended diagram is commutative, it follows from (3.5) that the following diagram

$$\begin{tikzcd}
H^{q-1}(G, C'') \arrow[r, "\delta"] & H^q(G, A'') \arrow[r, "\delta"] & H^{q+1}(G, A') \\
H^{q-1}(G, C'') \arrow[u, equal, "\mathrm{id}"] \arrow[d, equal, "\mathrm{id}"'] \arrow[r, "\delta"] & H^q(G, D) \arrow[u] \arrow[d] \arrow[r, "\delta"] & H^{q+1}(G, A') \arrow[u, equal, "\mathrm{id}"'] \arrow[d, equal, "-\mathrm{id}"] \\
H^{q-1}(G, C'') \arrow[r, "\delta"] & H^q(G, C') \arrow[r, "\delta"] & H^{q+1}(G, A'),
\end{tikzcd}$$

commutes as well, which immediately implies the theorem.